CHAPTER 4

BIOMOLECULES

CHAPTER ESSENCE AT A GLANCE:
Biochemical structure of macromolecules, protein folding hierarchies, and the catalytic kinetics of enzyme inhibition.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Primary and Secondary Metabolites:** Primary metabolites (amino acids, sugars, nucleotides, lipids) have direct, identifiable roles in normal physiological processes. Secondary metabolites (alkaloids like morphine, terpenoids like monoterpenes, toxins like abrin and ricin, lectins like concanavalin A, and drugs like vinblastine and curcumin) lack direct roles in primary metabolism but possess vital ecological and medical importance.

• **Protein Structural Hierarchy and Peptide Bonds:** Proteins are polymers of amino acids linked by peptide bonds. Primary structure represents the linear amino acid sequence (N-terminal to C-terminal); Secondary structure comprises alpha-helices and beta-pleated sheets stabilized by hydrogen bonds; Tertiary structure involves 3D folding into globular conformations creating active catalytic clefts; Quaternary structure arises from multi-subunit assemblies (e.g., hemoglobin with 2 alpha and 2 beta chains).

• **B-DNA Watson-Crick Architecture:** B-DNA is a right-handed antiparallel double helix: pitch is 3.4 nm (34 Angstroms), containing 10 base pairs per turn, with 0.34 nm spacing between adjacent base pairs. Purines pair with pyrimidines through complementary hydrogen bonds: Adenine pairs with Thymine via two hydrogen bonds, and Guanine pairs with Cytosine via three hydrogen bonds.

• **Enzyme Catalysis and Competitive Inhibition:** Enzymes lower the activation energy barrier without altering chemical equilibrium. In competitive inhibition, the inhibitor closely resembles the substrate in molecular structure and competes for the active site, increasing Km (lowering enzyme affinity) without changing Vmax (e.g., malonate inhibiting succinate dehydrogenase). Enzyme cofactors include prosthetic groups (tightly bound heme), coenzymes (NAD, NADP), and metal ions (Zn2+ in carbonic anhydrase).

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{B-DNA Parameters} = \text{Pitch: 3.4 nm}, \quad \text{10 bp per turn}, \quad \text{Distance: 0.34 nm}$$
Watson-Crick Double Helix Geometric Constants

$$\text{Competitive Inhibition} = K_m \uparrow \text{ (Affinity decreases)}, \quad V_{\max} = \text{Constant}$$
Michaelis-Menten Kinetic Shift under Competitive Inhibition

$$\text{Enzyme Co-factors} = \text{Apoenzyme (Protein)} + \text{Prosthetic Group, Coenzyme, or Metal Ion} = \text{Holoenzyme}$$
Holoenzyme Subunit Architecture and Catalytic Cofactors

$$\text{Succinate Dehydrogenase} + \text{Succinate} \rightleftharpoons \text{Fumarate} \quad [\text{Inhibited by Malonate}]$$
Classic Diagnostic Competitive Enzyme Inhibition Archetype

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In competitive inhibition, adding excess substrate overcomes the inhibitor, restoring the original reaction velocity; Km increases while Vmax remains unaltered.

• **Alert:** In B-DNA, the two polynucleotide chains run in opposite antiparallel directions: one chain runs 5' to 3' while the other runs 3' to 5'.

• **Alert:** Prosthetic groups are organic compounds permanently and tightly bound to the apoenzyme (e.g., heme in peroxidase), whereas coenzymes are transiently associated.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What are secondary metabolites? Give two examples of secondary metabolites used as therapeutic drugs. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Secondary metabolites are organic compounds produced by plants, fungi, and microbes that do not play an explicit, direct role in primary growth and development. **[1 Mark]**

• Therapeutic drug examples: Vinblastine (anticancer alkaloid) and Curcumin (anti-inflammatory compound from turmeric). **[1 Mark]**

2. Distinguish between prosthetic groups and coenzymes with a specific biochemical example for each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Prosthetic groups: Organic compounds that are tightly and permanently bound to the apoenzyme (e.g., Heme in catalase and peroxidase). **[1 Mark]**

• Coenzymes: Organic compounds whose association with the apoenzyme is transient, usually occurring during catalysis (e.g., NAD and NADP containing vitamin niacin). **[1 Mark]**

3. Explain the mechanism of competitive enzyme inhibition with reference to the inhibition of succinate dehydrogenase by malonic acid. How does it affect Km and Vmax? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Mechanism: Malonate closely resembles the substrate succinate in molecular structure and competes for the active catalytic site of succinate dehydrogenase. **[1 Mark]**

• When malonate binds the active site, it prevents succinate binding, decreasing the rate of conversion of succinate into fumarate. **[1 Mark]**

• Kinetic effect: Km increases (apparent substrate affinity decreases), while Vmax remains unchanged because high substrate concentrations displace the inhibitor. **[1 Mark]**

4. Describe the four levels of protein structure (primary, secondary, tertiary, and quaternary). At which level do enzymes gain biocatalytic activity? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Primary structure: Positional sequence of amino acids linked by peptide bonds, running from N-terminal to C-terminal. **[0.75 Mark]**

• Secondary structure: Local folding of polypeptide chains into alpha-helices and beta-pleated sheets stabilized by hydrogen bonds. **[0.75 Mark]**

• Tertiary structure: Overall 3D globular folding stabilized by disulfide, ionic, hydrophobic, and van der Waals interactions; creates catalytic active site clefts where enzymes gain biocatalytic activity. **[0.75 Mark]**

• Quaternary structure: Spatial arrangement of multiple polypeptide subunits relative to each other (e.g., adult hemoglobin with 2 alpha and 2 beta chains). **[0.75 Mark]**

5. State the key structural dimensions of the Watson-Crick B-DNA double helix regarding pitch, base pairs per turn, strand polarity, and complementary hydrogen bonds. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Pitch and base pairs: The pitch of the B-DNA helix is 3.4 nm (34 Angstroms), containing approximately 10 base pairs per complete turn, with 0.34 nm distance between adjacent base pairs. **[1 Mark]**

• Strand polarity: The two polynucleotide chains possess antiparallel polarity: one strand runs 5' to 3' and the opposite strand runs 3' to 5'. **[1 Mark]**

• Base pairing: Adenine pairs exclusively with Thymine through two hydrogen bonds (A = T); Guanine pairs with Cytosine through three hydrogen bonds (G = C). **[1 Mark]**

6. Explain how enzymes increase the rate of biochemical reactions by lowering the activation energy barrier. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Activation energy is the minimum additional energy required by substrate molecules to reach the unstable, high-energy transition state before forming products. **[1 Mark]**

• Enzymes bind substrate molecules at their active site to form an enzyme-substrate (ES) complex, straining bonds and stabilizing the transition state. **[1 Mark]**

• This significantly lowers the activation energy barrier, allowing a much higher fraction of substrate molecules to overcome the barrier per second, enhancing reaction velocity by millions of times. **[1 Mark]**

7. An enzymatic assay reveals that increasing substrate concentration gradually reverses the inhibitory effect of a newly synthesized chemical agent. Identify the type of enzyme inhibition, explain the molecular basis of this behavior, and predict the effect on Km and Vmax. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Inhibition type: Competitive enzyme inhibition. **[1 Mark]**

• Molecular basis: The inhibitor molecule shares close structural resemblance with the natural substrate, binding reversibly to the active catalytic site. **[1.5 Marks]**

• Reversal by excess substrate: When substrate concentration is elevated, substrate molecules statistically outcompete inhibitor molecules for active site occupancy, restoring full catalytic speed. **[0.5 Mark]**

• Effect on kinetics: Km increases (affinity of enzyme for substrate decreases), while Vmax remains unchanged. **[1 Mark]**